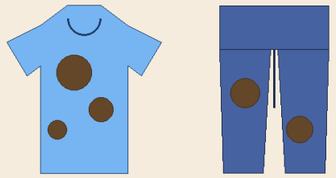
LESSON 4 · ROUND PROGRESSION

Manual to Batch Processing in 5 Steps

Each round introduces one computing concept: manual file handling, wildcards, loops, remote transfer and scheduled jobs

ROUND 1

Local Washer



3 files · round1_local/

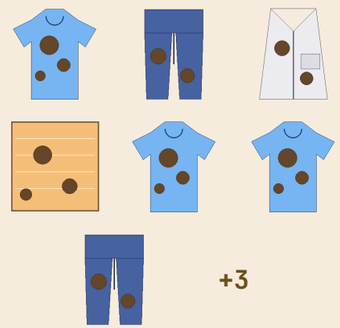
CONCEPT · MANUAL FILE HANDLING

```
mv dirty_laundry/round1_local/
   shirt_dirty.png washer/
bash scripts/local_washer.sh
```

Pipeline: input → process → output → verify.

ROUND 2

Too Much Laundry



11 files · round2_big_load/

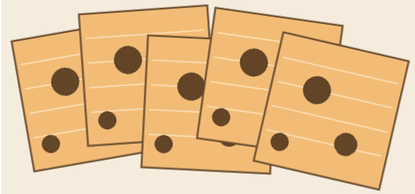
CONCEPT · WILDCARD PATTERNS

```
mv dirty_laundry/round2_big_load/
   *_dirty.png washer/
```

One pattern selects many files at once.

ROUND 3

Robot Washer Loop



5 files · round3_loop_challenge/

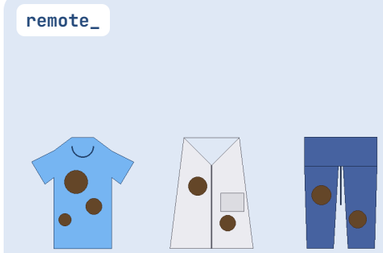
CONCEPT · LOOPS

```
bash scripts/robot_move_to_washer.sh \
  dirty_laundry/round3_loop_challenge
```

A loop applies the same operation to every file.

ROUND 4

PI Laundromat



5 files · round4_remote_load/

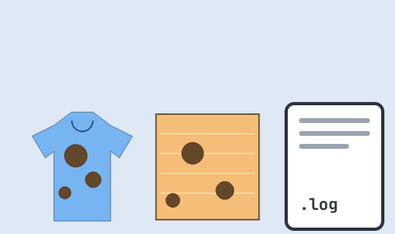
CONCEPT · REMOTE ACCESS AND TRANSFER

```
bash scripts/fake_ssh.sh \
  student@pi-laundromat
bash scripts/fake_scp.sh \
  dirty_laundry/round4_remote_load \
  remote_laundromat/incoming
```

Data must be transferred before remote processing.

ROUND 5

Laundry Cluster Job



5 files · round5_cluster_job/

CONCEPT · JOB SUBMISSION AND LOGS

```
bash scripts/submit_laundry_job.sh
ls remote_laundromat/logs
cat …/logs/latest_job.log
```

Jobs run unattended; logs record what happened.

ON THE LOCAL MACHINE

increasing automation →

ON A REMOTE SERVER (SIMULATED)